Rheinisch-Westfälische Technische Hochschule Aachen

Institut für physikalische Chemie

A machine learning model for the prediction of hessian matrices for two to four atomic molecules.

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Research Internship Report

submitted by

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List of abbreviations

APF	Atom Pair Focused
BO	Born–Oppenheimer
CS	Coordinate System
ETR	Extra Tree Regression
ES	Electrostatic
LDA	Local Density Approximation
MI	Main Inertial
ML	Machine Learning
RFR	Random Forest Regression
XC	Exchange and Correlation

1 Aim

2 Motivation

5 Results and Discussion

5.1 The ML Models and Data Set

For every prediction of the hessian matrix, it is necessary to use the homonuclear and heteronuclear model. Both models were further investigated by three adjustments, namely the algorithms, the labeling and a generalization.
At first, as described in the Chapter Theory, the RFR and ETR are used as the different algorithms for the fitting of the ML model. For the labeling the two possible cases of the labeling, which are the indexed and permuted case were discussed. These are abbreviated with Idx(Indexed) and Perm(Permuted). At last, it is investigated, how the performance changes if instead of fitting the ML model only case–specific the ML model is fitted to a general model, that predicts all cases. For this differentiation, the abbreviations Case (case–specific) and Gen (general) are used. As an example, the abbreviation for the model with the ETR as the algorithm, the indexed labeling and general case is ETR–Gen–Idx.
A data set for benchmarking the ML model was constructed. The set consists of eight different molecules. These are namely H_2, N_2, O_2, F_2, HF, CO, H_2O, and NH_3. For each molecule, structures with different bond length were generated. Only the distance of one bond was changed for each molecule. This lead into 103 different structures. These structures were split into a training set of 75% and a test set of 25%. The split is done structure wise to achieve the possibility to compare other quantities then the hessian elements. For each structure, the hessian and feature were calculated with the GFN2–xTB method.

5.2 The H_2 system

As a first approach, the ETR–Case–Perm model was used for the subset of the hydrogen molecule. For this the wave numbers $\tilde{\nu}_{\mathrm{ML}}$ were predicted and plotted against the wave numbers $\tilde{\nu}_{\mathrm{xTB}}$ calculated via GFN2–xTB. This is done for 5 random states and is shown in Fig. 5.1. The black line is the diagonal, on which the data points would appear for a perfect prediction. Furthermore, the wave number increases with the decrease of the bond length. For the approximated range of $1800\ \mathrm{cm}^{-1} < \tilde{\nu}_{\mathrm{xTB}} < 7500\ \mathrm{cm}^{-1}$ there are abbreviations from the diagonal and therefore a sufficient prediction. In respect to the small data set of only eleven structures for the full subset of hydrogen, the prediction is

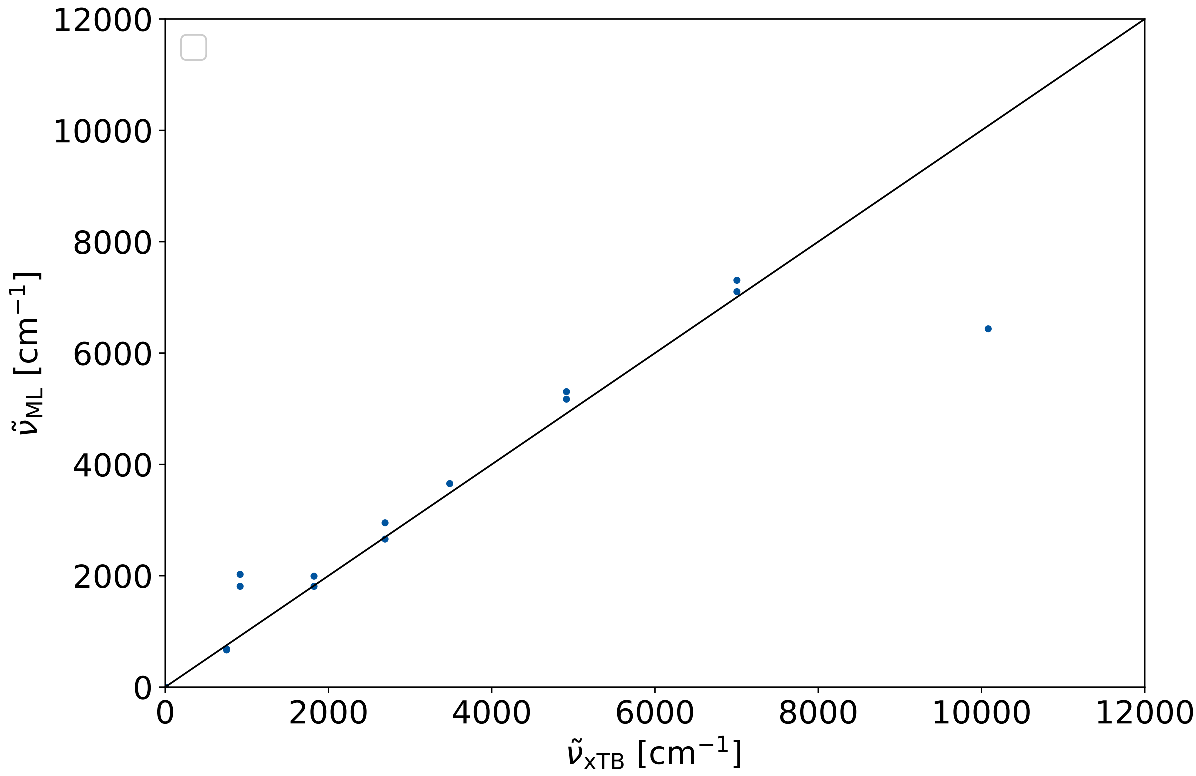

Figure 5.1: The wavenumber, predicted via the ML model $\tilde{\nu}_{\mathrm{ML}}$ is plotted against the wavenumber calculated via GFN2–xTB $\tilde{\nu}_{\mathrm{xTB}}$. The model uses the Extra Tree Regression. The sample set is H_2 and the training set is 75 % of the whole set. The prediction were done for 5 random states.

The abbreviations are stronger in the cases of 1800 $\mathrm{cm}^{-1} > \tilde{\nu}_{\mathrm{xTB}}$ and $\tilde{\nu}_{\mathrm{xTB}} > 7500\ \mathrm{cm}^{-1}$. These predictions are probably done for cases, where the training set does not include molecules, which have a lower or higher distance, respectively. This assumption is underlined by the fact, that the predicted values correspond to the next higher or lower value. For example, the next lowest value in the case of the prediction at $\tilde{\nu}_{\mathrm{xTB}} \approx 1 \cdot 10^5 \mathrm{cm}^{-1}$ is $\tilde{\nu}_{\mathrm{xTB}} \approx 7 \cdot 10^4 \mathrm{cm}^{-1}$. This in the same order as the predicted value $\tilde{\nu}_{\mathrm{ML}} \approx 6.4 \cdot 10^4 \mathrm{cm}^{-1}$ at $\tilde{\nu}_{\mathrm{xTB}} \approx 1 \cdot 10^5 \mathrm{cm}^{-1}$. This matches to the expected behavior for a decision tree based algorithms (cf. Fig. 3.2).
This concludes in the sufficient description of the wave numbers for interpolation. It could also be shown, that the prediction fails for extrapolation.

5.3 Comparison of RFR and ETR

For the comparison of the algorithms, the RFR–Gen-Perm and ETR–Gen–Perm models were used. These models are compared to each other by testing there dependency on the amount of training set used. As the quantity for testing the performance the MSE between the predicted

hessian and the hessian computed via GFN2–xTB is used. The MSE was calculated for ten different random states and the mean of these MSE was calculated.
It is expected, that the performance of the ML model should increase, the more data it is trained with. To verify this behavior, the training set was increased successively. The full training set is 75 % of the full data set. Thus, test set is 25 %. While keeping the test set constant, the training set is changed successively between 7.5 % of the data set to 75 %. Fig 5.2 shows the MSE in the dependency of the amount of training set used for the ETR (on the left) and the RFR (on the right). The error bars indicate the standard deviation.

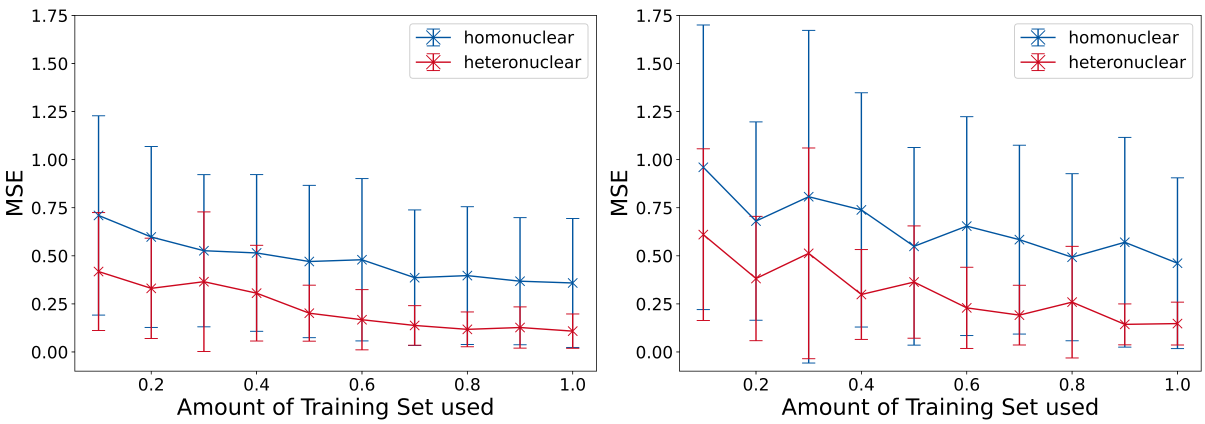

Figure 5.2: Learning plots for the predicting hessian elements $\frac{\partial^2 E}{\partial a \partial b}$ for an increase of the amount of training set used. On the left is the learning plot of the Extra Tree Regression and on the right the plot of the Random Forest Regression shown. The full training set is 75 % of the sample set. The full sample set, excluding the H_3 system, is used. The mean of ten random states were used for the MSE. The plots are done for the homonuclear model (blue) and heteronuclear model (red).

In the case of the ETR, the MSE decreases with an increase of the amount of training set used for the homonuclear and heteronuclear model. This behavior is also recognizable in the case of the RFR. In comparison to the ETR, the RFR shows a higher standard deviation for the data points. Thus, the RFR depends more strongly on the random state than the ETR. Furthermore, the ETR has, especially for the homonuclear model, lower MSE values for any amount of training set. At the full amount of training set and at 90% of the training set, the RFR and ETR show a similar performance for the heteronuclear model. All in all, the ETR shows a better performance for the whole investigated range.
There is the possibility, that the ML model has so much information in the features, that it can always predict the target. Hence, the model is tested on whether this is the case, or the ML model learns from the information of the feature. For this, the features are scattered with a random factor. The intensity of the scattering is amplified by the factor σ. It is expected, that a model that learns from the features performs worse, if the scattering of the data points

increases.

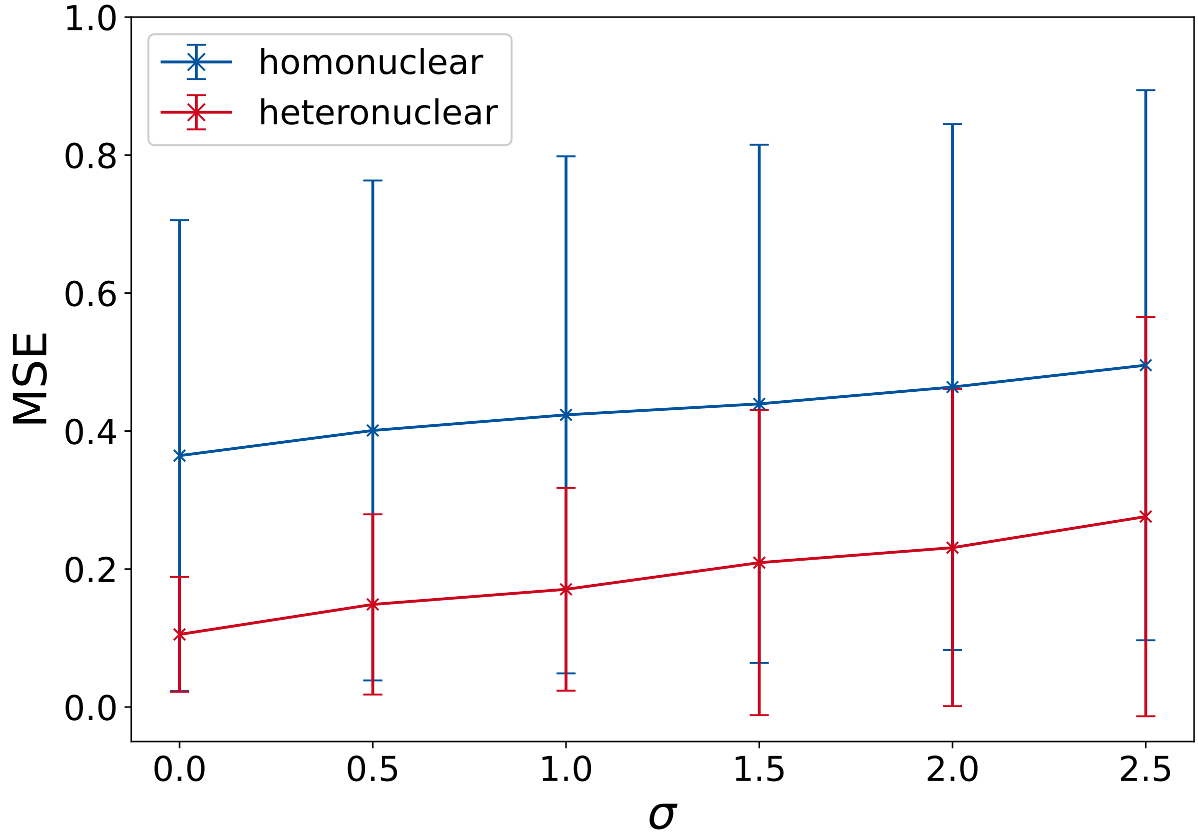

Figure 5.3: Learning plots for predicting hessian elements $\frac{\partial^2 E}{\partial a \partial b}$ for a variation of the factor σ. The full sample set, excluding the H_3 system, is used for the Ml models. The mean of ten random states were used for the MSE. The plots are done for the homonuclear model (blue) and heteronuclear model (red).

The performance of the model ETR–Multi–Permuted was tested for this. The prediction was done for ten different random states. The MSE between the predicted hessian elements and the via GFN2–xTB calculated ones was investigated. For the investigated range of $0 \leq \sigma \leq 2.5$ the MSE increases monotonically. Hence, this shows, that the model learns from the features and is not purely random.

In this section a comparison of the RFR and ETR was done. The ETR shows overall a better performance than the RFR. Furthermore, it could be verified, that the ML model learns statistically from the features used for the prediction of the hessian matrices.

5.4 Comparison of Permutation Model and Indexed Model

Beside the different algorithms for the construction of the trees, the Case and Gen as also Idx and Perm models were compared to another. The models ETR–Gen–Perm, ETR–Gen–Idx, ETR–Case–Perm and ETR–Case–Idx are investigated. The quantities for measuring are the hessian elements, the wave numbers and the zero point vibrational energy. The difference between the predicted and via GFN2–xTB computed quantities were computed. From these differences the standard deviation as also the mean were calculated, and a Gaussian distribution was assumed. At first the direct prediction of the hessian elements were investigated, and its distribution is shown in Fig 5.4.

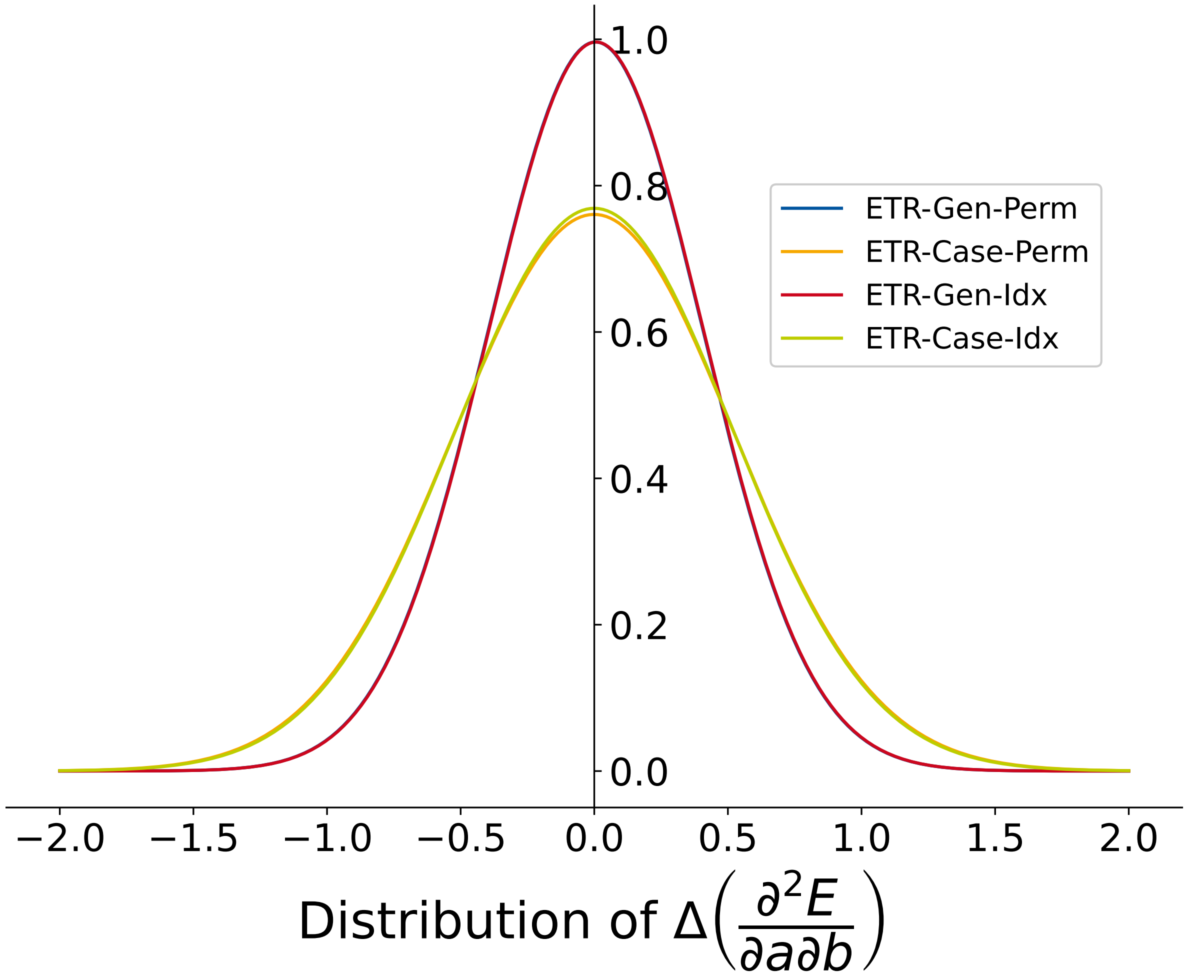

Figure 5.4: Distribution of $\Delta\left(\frac{\partial^2 E}{\partial a \partial b}\right)$ for different models. In every model, the Extra Tree Regression was used. The distribution is assumed to be Gaussian type. For this the median and standard deviation of $\Delta\left(\frac{\partial^2 E}{\partial a \partial b}\right)$ are computed.

Comparing the ETR–Case–Perm and ETR–Case–Idx, as also ETR–Gen–Perm and ETR–Gen–Idx shows, that both approaches for the labeling of the features do not differ significantly. Thus, both approaches are in good agreement with another. This shows, on the one hand, the strength of the decision tree algorithms. They cannot overfit and seem to be capable of describing this complex system with a simple differentiation by indexes. On the other hand, the permuted model does the same prediction for the model, but may be useful for other machine learning algorithms or even neural networks.

The comparison of the ETR–Case–Perm and ETR–Gen–Perm shows, that the ETR–Case–Perm model deviates more strongly. This is in agreement with the comparison of the ETR–Case–Idx and ETR–Gen–Idx models. Though the deviation is stronger, the means of the models do not differ significantly. The data set for training the models is smaller for the case–specific models than for the general models. This might be the reason for the deviation.

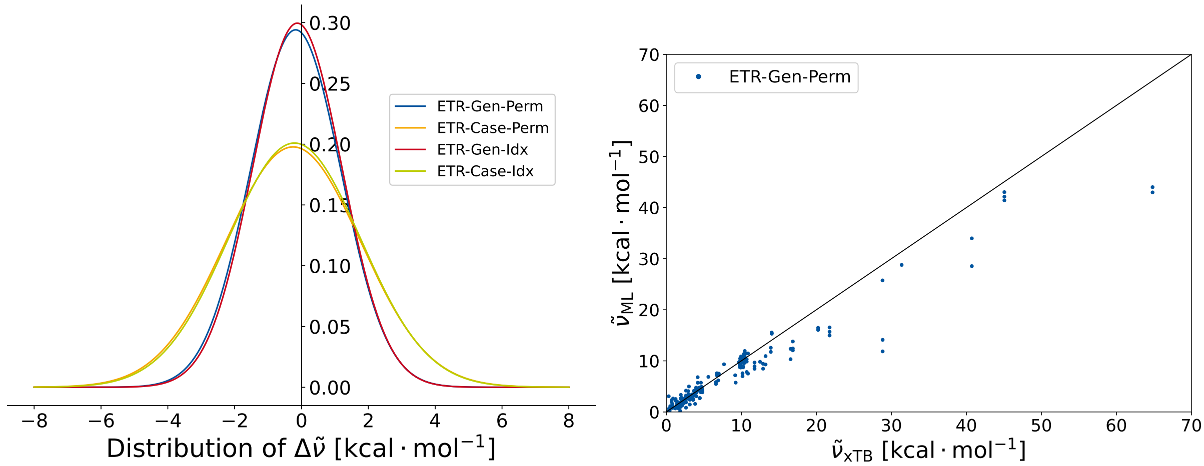

Figure 5.5: On the left is the distribution of $\Delta\tilde{\nu}$ for different models. In every model, the Extra Tree Regression was used. The distribution is assumed to be Gaussian type. For this the median and standard deviation of $\Delta\tilde{\nu}$ are computed. On the right is the $\tilde{\nu}_{\mathrm{ML}}$ plotted against $\tilde{\nu}_{\mathrm{xTB}}$. The wave numbers are predicted with the ETR–Gen–Perm model.

The wave numbers $\tilde{\nu}$ of the harmonic frequencies of the molecules were calculated. The difference between the difference between the ML predicted $\tilde{\nu}_{\mathrm{ML}}$ and the via GFN2–xTB computed $\tilde{\nu}_{\mathrm{xTB}}$ was calculated. A Gaussian distribution was assumed. This is plotted in Fig. 5.5 for the four different varieties of the model. Furthermore, the plot of the $\tilde{\nu}_{\mathrm{ML}}$ vs. $\tilde{\nu}_{\mathrm{xTB}}$ is shown. The relative distributions of the four variations of the models are similar to the ones of the hessian elements. The ETR–Gen–Perm and ETR–Gen–Idx show a better performance than the ETR–Case–Perm and the ETR–Case–Idx. The mean is shifted to a negative $\Delta\tilde{\nu}$. In contrast to the Gaussian distribution of the hessian elements, the Idx models show a marginal better performance than the Perm model. The deviations of several $\mathrm{kcal}\cdot\mathrm{mol}^{-1}$ are rather strong. These strong deviations might be due to a high extrapolation errors, which lead to high standard distribution For example at $\tilde{\nu}_{\mathrm{xTB}} \approx 65\mathrm{kcal}\cdot\mathrm{mol}^{-1}$ the deviation might be due to extrapolation. Hence, the difference of about $\Delta\tilde{\nu} \approx 20\mathrm{kcal}\cdot\mathrm{mol}^{-1}$ has a strong influence on the distribution, although ML is not customized for extrapolation problems.

5.5 Benchmarking the H_3 system

Especially, the prediction of hessian matrices of fully unknown systems is of interest. Hence, the ETR-Gen-Perm and ETR-Gen-Idx models were tested on the performance on a new, for the model unknown, system. For the H_3 system the outer hydrogen atoms were fixed, while the hydrogen atom was shift from one hydrogen to another. For each position of the hydrogen atom, the force constants of the systems were predict via the ML model. The force constants were also calculated via GFN2–xTB.

Fig. 5.6 shows the force constants calculated by GFN2–xTB and predicted via the ETR–Gen–Idx model. Interesting to point out is the symmetric behavior of the system. The ML model should, in principle, mimic this behavior.

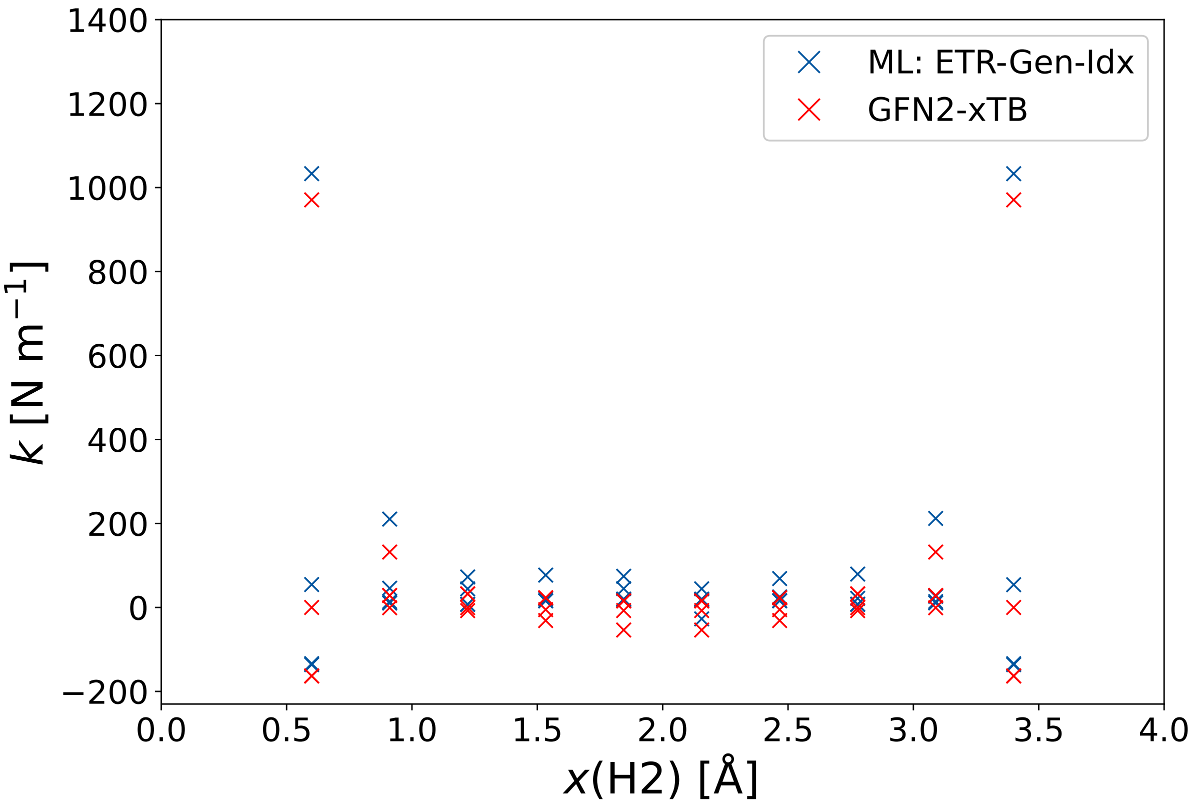

Figure 5.6: Force constants predicted for the H_3 systems in dependence of the distance between H1 and H2 are shown. The force constants are shown for the GFN2–xTB calculation (red), as also the via the ML model predicted values (blue). The full sample set, excluding the H_3 system, was used for the ML model. The ML model ETR–Gen–Idx was used for the prediction.

While this is overall the case, there are abbreviation from this behavior. For the example, the symmetry is not given in the case of $x(\mathrm{H2}) = 1.84$ Å and $x(\mathrm{H2}) = 2.15$ Å.

Fig. 5.7 shows the prediction of the focre constants via the ETR–Gen–Perm model. Similar to the ETR–Gen–Idx model, the symmetry is in general achieved. Nevertheless, for example the force constants at $x(\mathrm{H2}) = 0.91$ Å and $x(\mathrm{H2}) = 3.08$ Å show abbreviations from oneanother.

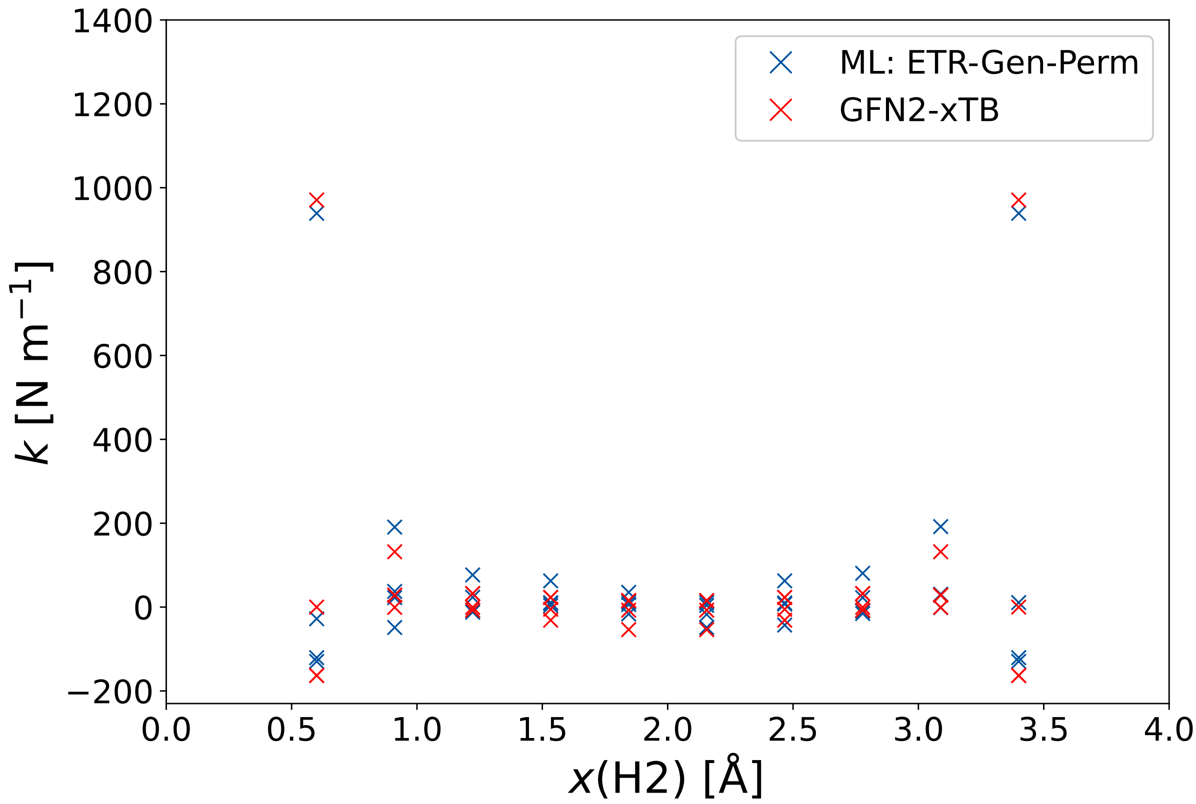

Figure 5.7: Force constants predicted for the H_3 systems in dependence of the distance between H1 and H2 are shown. The force constants are shown for the GFN2–xTB calculation (red), as also the via the ML model predicted values (blue). The full sample set, excluding the H_3 system, was used for the ML model. The ML model ETR–Gen–Perm was used for the prediction.

Further investigations on the models are necessary, to locate the reason for the break of symmetry. Worth considering are the rotation process, the homonuclear and heteronuclear model. The rotation process can be neglected, as rotation is also done for the GFN2–xTB calculations. A decoupled investigation of the homonuclear and heteronuclear model is therefore necessary.

6 Conclusion & Outlook

The goal of the research was the prediction of hessian matrices by a machine learning model. The machine learning model was based on atomic features and the prediction of the elements of the hessian matrices was done atom pair wise. The model was benchmarked by a small data set of 8 molecules with 103 different structures. The computations of the hessian matrices and features for these systems was done with the GFN2–xTB method.
In first place, the general learning of the model was investigated. A statistical learning behavior of the ML models RFR–Gen–Perm and ETR–Gen–Perm could be observed. Compared to another, the ETR–Gen–Perm model showed a better performance. Furthermore, a physical correlation could be presented for the ETR–Gen–Perm model.
Moreover, 4 variations of the model were investigated and compared to another, which are the models ETR–Gen/Case–Perm/Idx. Here a similar behavior for different labeling approaches (Indexed and Permuted) was shown. The general models show a better prediction than the case–specific models. Here, the hessian elements, zero point vibrational energy and wave numbers were predicted.
In addition, the set was used for the prediction of a system, that is unknown to the ML model. The data set of the new H_3 system shows a symmetric behavior for the force constants, which could, with some minor errors, be depicted by the ML models.

The next necessary approaches for this research, is the improvement of the models in there symmetric behavior. Although, for the H_3 system the errors in symmetry are minor, the adjustment to a full symmetric behavior is mandatory. In addition, a set of 103 structures is small for a ML model. Hence, the date set size should be increased and tests on benchmark data sets should be done for a comparison to literature. When the prediction of the hessian matrix on GFN2–xTB level is sufficient, the model can be adapted to a ΔML model.

7 Appendix

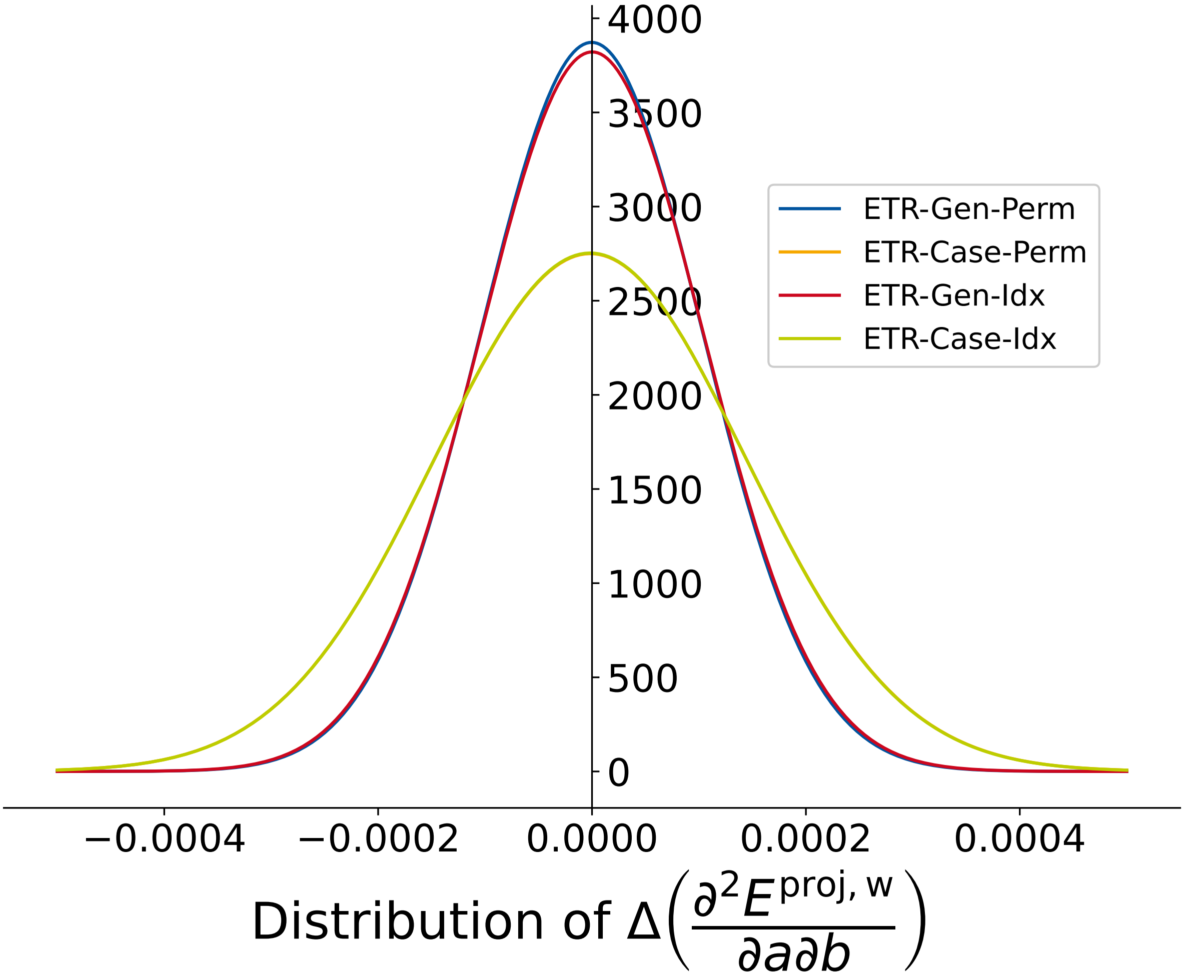

Figure 7.1: Distribution of $\Delta\left(\frac{\partial^2 E^{\text{proj,w}}}{\partial a \partial b}\right)$ for different models. In every model, the Extra Tree Regression was used. The distribution is assumed to be Gaussian type. For this the median and standard deviation of $\Delta\left(\frac{\partial^2 E^{\text{proj,w}}}{\partial a \partial b}\right)$ are computed.

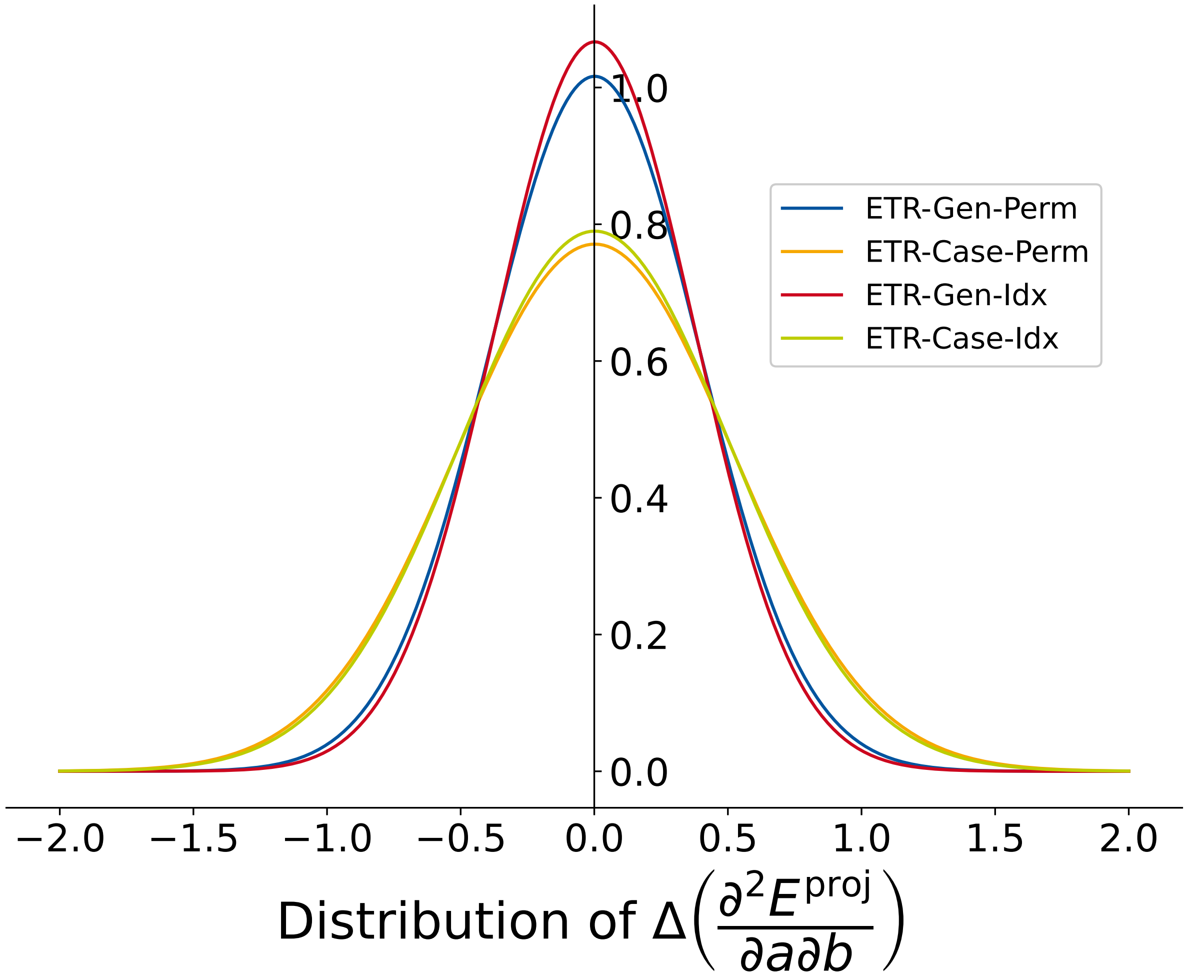

Figure 7.2: Distribution of $\Delta\left(\frac{\partial^2 E^{\mathrm{proj}}}{\partial a \partial b}\right)$ for different models. In every model, the Extra Tree Regression was used. The distribution is assumed to be Gaussian type. For this the median and standard deviation of $\Delta\left(\frac{\partial^2 E^{\mathrm{proj}}}{\partial a \partial b}\right)$ are computed.

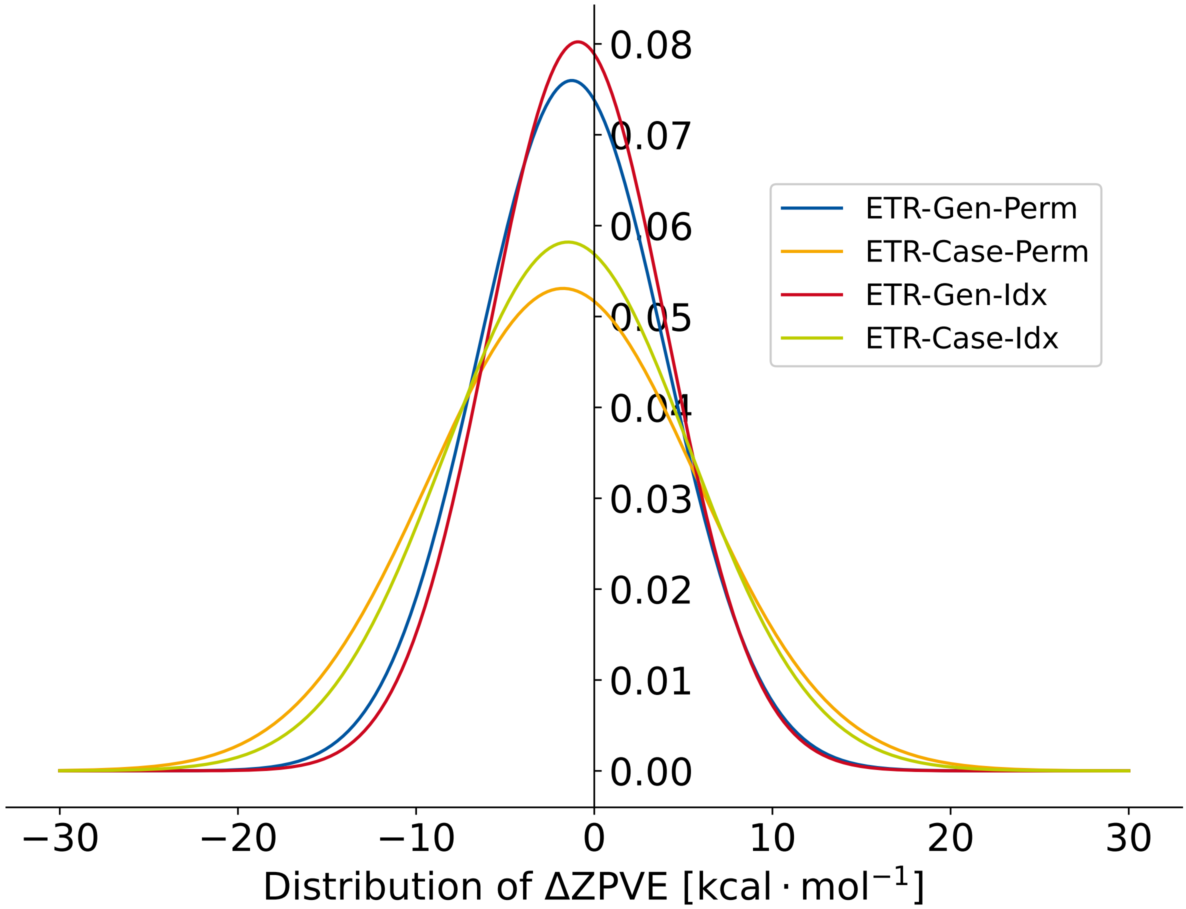

Figure 7.3: Distribution of ΔZPE for different models. In every model, the Extra Tree Regression was used. The distribution is assumed to be Gaussian type. For this the median and standard deviation of ΔZPE are computed.

$$CN_A = \sum_{B \neq A}^{N_{\text{atoms}}} \frac{1}{1 + \exp\left(\frac{-16a \cdot 4(R_{A,\text{cov}} + R_{B,\text{cov}})}{3R_{AB} - 1}\right)} \tag{7.0.1a}$$

$$CN^{\text{ext}} = \sum_{B \neq A}^{N_{\text{atoms}}} \frac{CN_B}{f_{\text{damp}}(R_A, R_B)} \tag{7.0.1b}$$

$$q_A = Zg_A - \sum_{\mu \in A}^{N_{\text{BF}}} \sum_{\nu}^{N_{\text{BF}}} P_{\mu\nu} S_{\mu\nu} \tag{7.0.1c}$$

$$q_A^{\text{ext}} = q_A + \sum_{B \neq A}^{N_{\text{atoms}}} \frac{q_B}{f_{\text{damp}}} \tag{7.0.1d}$$

8 Bibliography